CFAT 1.1.1

Types of data and displaying distributions

1A, 1B, 1C, 1D, 1E · CAS and bound reference allowed

Name _____

/13

SECTION A · MULTIPLE CHOICE

Circle one answer. 6 marks

QUESTION 1 — 1 mark

The variables *number of moths* (less than 250, 250-500, more than 500) and *trap type* sugar, scent, light) are…

A. both nominal variables.
B. both ordinal variables.
C. a numerical variable and a categorical variable respectively.
D. a nominal variable and an ordinal variable respectively.
E. an ordinal variable and a nominal variable respectively.

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QUESTION 2 — 1 mark

Data was collected on insects caught in a trap. The variables were: *colour*, *name of species*, *number of wings*, *body length* (mm) and *body weight* (mg).

How many of these variables are discrete?

A. 1
B. 2
C. 3
D. 4
E. 5

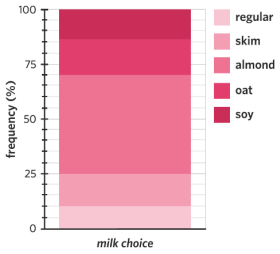
Edrolo U3&4 1A Q15

QUESTION 3 — 1 mark

The percentage segmented bar chart shows the type of milk customers ordered with their coffee. 36 people ordered almond milk.

The number of people who ordered regular milk is closest to

A. 8
B. 10
C. 15
D. 25



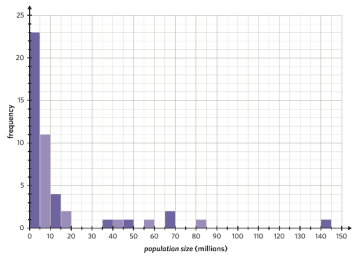
Edrolo U3&4 1B Q9

QUESTION 4 — 1 mark

The histogram shows the *population size* of 48 countries in 2018.

The number of countries with a population size between 5 million and 20 million is

A. 11
B. 17
C. 23
D. 34
E. 35



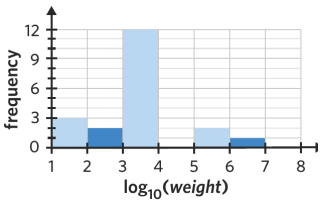
Edrolo U3&4 1C Q17

QUESTION 5 — 1 mark

The histogram shows the *weight*, in grams, of 20 animal species, plotted on a $\log_{10}$ scale.

The percentage of these species with a weight of less than 10 000 g is

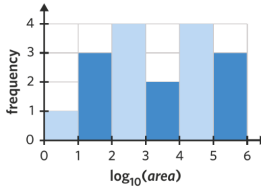
A. 17%
B. 70%
C. 75%
D. 80%
E. 85%



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QUESTION 6 — 1 mark

The histogram shows $\log_{10}(area)$, with *area* in km², for 17 islands.



The modal area of these islands, in km², is between

A. 2 and 3
B. 2 and 3, as well as 4 and 5
C. 2 and 5
D. 10 000 and 100 000
E. 100 and 1000, as well as 10 000 and 100 000

Edrolo U3&4 1D Q14

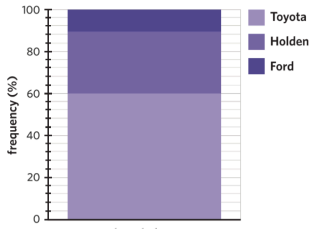
SECTION B · SHORT ANSWER

Show your working. 7 marks

QUESTION 7

2 marks

The brands of 50 cars entering a carpark were recorded, with the results shown in the percentage segmented bar chart provided.



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a Write a report to describe the distribution of cars entering the carpark, being sure to reference the modal category.

QUESTION 8

3 marks

Simon conducted a survey for his statistics class by asking 25 of his classmates how long it took them to get to school on a particular day. He displayed his findings in a stem plot.

Key: 0 | 5 = 5 minutes

0	5 5 6
1	2 3 6 8 9
2	1 4 5 5 6 7 9
3	2 3 4 8 8 8
4	0 1 6
5	2

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a How many students took between 20 and 30 minutes to get to school?

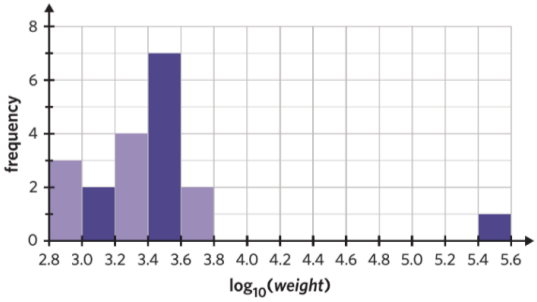
b How many students took more than 25 minutes to get to school?

c What is the mode?

QUESTION 9

2 marks

The approximate weight of different vehicles (cars, boats, planes) was recorded in kilograms. The data was displayed by a histogram using the log scale.



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a What weight is represented by 3.6 on the log scale, correct to the nearest kg?

b A truck is known to have a weight of 3750kg. In which interval will this value be plotted?